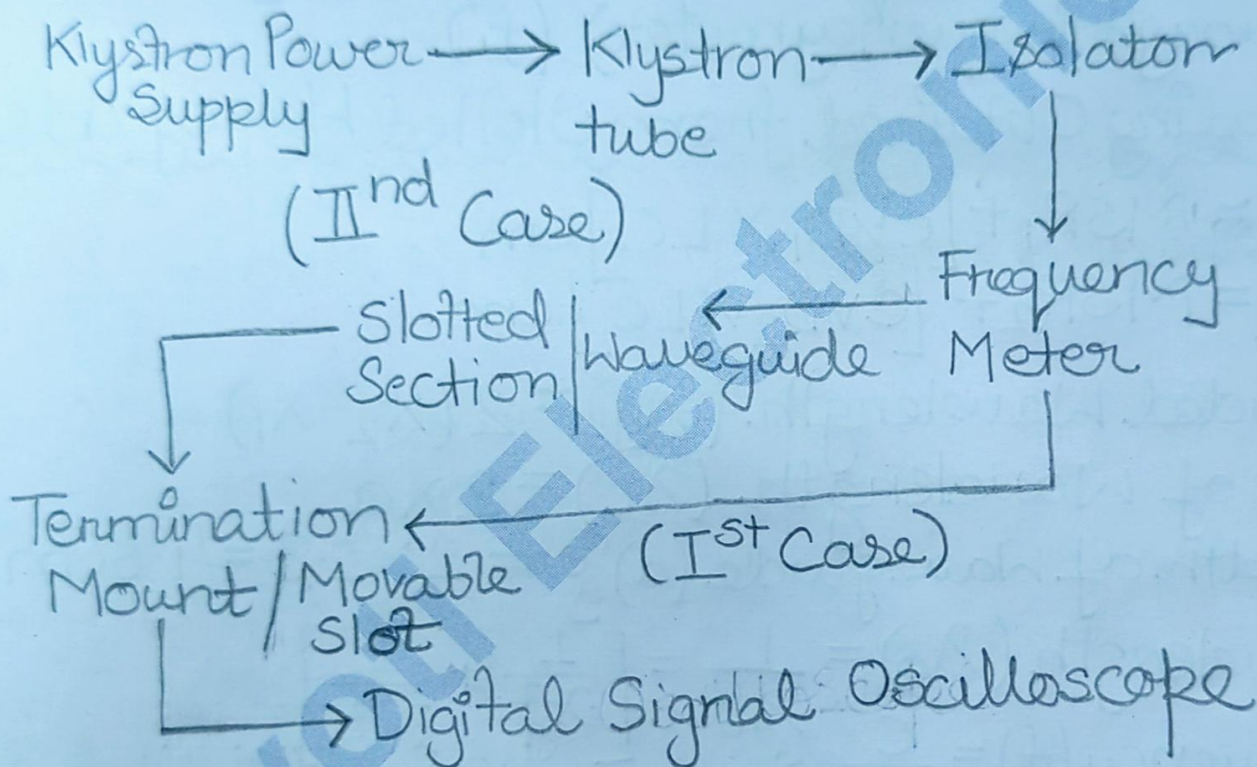


25.08.25 Experiment No-3

* Objective :- Determination of VSWR and Reflection Coefficient

* Experiment Set-up



* Klystron Power Supply Setup

1. Modulation AM
2. Frequency Maximum
3. Amplitude Maximum
4. Beam Voltage Minimum
5. Repeller Voltage Maximum.

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*Initial Parameters

1. Beam Voltage : 234 Volts $\approx$ 260 Volts
2. Beam Current : 10 milli Amp $\approx$ 10 milli amp
3. Repeller Voltage : 269 Volts (maximum)
4. Measured Frequency : 9.595 GHz
(through frequency meter) (f_1)

*Reading Obtained from Slotted Waveguide

$$\Rightarrow X_1 = MSR_1 + [CVD_1 \times LC] \text{ cm}$$

$$\Rightarrow X_2 = MSR_2 + [CVD_2 \times LC] \text{ cm}$$

$$\text{Guided Wavelength } (\lambda_g) = 2(X_2 - X_1)$$

$$\text{Cut off Wavelength } (\lambda_c) = 2 \times a$$

$$\{\text{Width of Waveguide } (a)\} = 2 \times 2.3 = 4.6 \text{ cm}$$

$$\text{Wavelength } (\lambda_0) = \frac{1}{\frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}}$$

$$\text{Frequency } (f) = \frac{c}{\lambda_0} \quad \{c = 3 \times 10^{10} \text{ cm/sec}\}$$

* Experiment Procedure $\{LC = 0.01\}$

$$\begin{array}{l} MSR_1 = 11.3 \\ CVD_1 = 4 \\ MSR_2 = 13.2 \\ CVD_2 = 5 \end{array} \left. \begin{array}{l} \text{Maxima 1} \\ \\ \text{Maxima 2} \end{array} \right\} \text{Reading from Slotted Section}$$

$$\Rightarrow X_1 = 11.3 + [4 \times 0.01] = 11.34 \text{ cm}$$

$$\Rightarrow X_2 = 13.2 + [5 \times 0.01] = 13.25 \text{ cm}$$

$$\lambda_2 - \lambda_1 = 3.82 \text{ cm}$$

$$2 \times 2.3 = 4.6 \text{ cm}$$

$$\frac{1}{\lambda_0^2} = \left[\frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2} \right]$$

$$\Rightarrow \frac{1}{\lambda_0^2} = \left[\frac{1}{(3.82)^2} + \frac{1}{(4.6)^2} \right]$$

$$\Rightarrow \frac{1}{\lambda_0^2} = \frac{1}{14.59} + \frac{1}{21.16}$$

$$\Rightarrow \frac{1}{\lambda_0^2} = 0.1157878024$$

$$\Rightarrow \lambda_0^2 = 8.636488294$$

$$\Rightarrow \lambda_0 = 2.93 \text{ cm}$$

$$\Rightarrow f_2 = \frac{c}{\lambda_0} = \frac{3 \times 10^{10}}{2.93} = 1.02 \times 10^{10} = 10.20 \text{ GHz}$$

$$\Rightarrow f_2 - f_1 = 10.20 - 9.595 \text{ GHz}$$

$$= 0.605 \text{ GHz}$$

24/08/25
* VSWR

$$V_{\max} = 5.20 \text{ mV}$$

$$V_{\min} = 3.68 \text{ mV}$$

$$VSWR = \frac{V_{\max}}{V_{\min}} = \frac{5.20}{3.68} = 1.41$$

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$$\Rightarrow \sqrt{1.41} = 1.187 \leftarrow$$

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* Reflection Coefficient

$$\Rightarrow S_{11} = \frac{VSWR - 1}{VSWR + 1}$$

$$= \frac{1.41 - 1}{1.41 + 1}$$

$$= \frac{0.41}{2.41} = 0.17$$

linear to dB scale

$$\Rightarrow 20 \log_{10} (0.17)$$

$$= -15.39 \text{ dB}$$

* Anything that is below -10 dB or $|S_{11}| < -10 \text{ dB}$ is overdriving or have transmitted 90% of input power.

Why? & How to justify it?

$$\Rightarrow P_t = P_{in} (1 - |\Gamma|^2)$$

P_t is transmitted power

P_{in} is input power

Γ is reflection coefficient

$\Rightarrow P_t = 90\%$; in term of P_{in} it would be

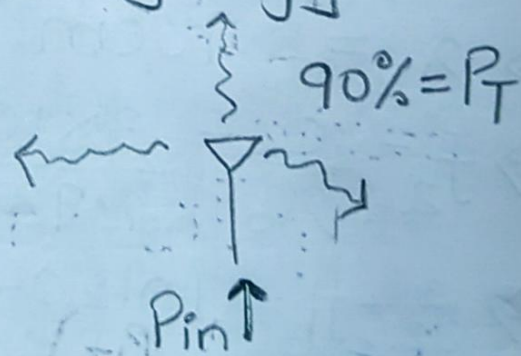
$$\Rightarrow P_t = 0.9 P_{in}$$

$$\Rightarrow 0.9 P_{in} = P_{in} (1 - |\Gamma|^2)$$

$$\Rightarrow 0.9 = 1 - |\Gamma|^2$$

$$\Rightarrow \Gamma = 0.316$$

$$\Rightarrow S_{11} = -20 \log_{10} \left| \frac{1}{\Gamma} \right| = -20 \log_{10} \frac{1}{0.316} = -10 \text{ dB}$$



$$-10.00625835$$

$$-10.00625835$$

$$\text{B.S.} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + |0.316|}{1 - |0.316|} = 1.92 \approx 2$$

Reflected Power (%)

$$\Rightarrow |\Gamma|^2 \times 100 = |0.316|^2 \times 100$$

$$= 0.099 \times 100$$

$$= 9.9856\%$$

Relation Between VSWR and Reflection Coefficient

$$\Rightarrow v = v_1^+ e^{-j\beta z} + v_1^- e^{+j\beta z}$$

This equation is in phasor and we need to take it to time domain

$$\begin{array}{ll} v_1^+ e^{-j\beta z} & v_1^- e^{+j\beta z} \\ \text{Re}(v_1^+ e^{-j\beta z} e^{j\omega t}) & \text{Re}(v_1^- e^{+j\beta z} e^{j\omega t}) \\ v_1^+ \cos(\omega t - \beta z) & v_1^- \cos(\omega t + \beta z) \end{array}$$

$$\Rightarrow v = v_1^+ e^{-j\beta z} + v_1^- e^{+j\beta z}$$

We know;

$$\Rightarrow \Gamma = \frac{v_1^-}{v_1^+}; \Rightarrow v_1^- = \Gamma v_1^+$$

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$$\Rightarrow v = v_1^+ e^{-j\beta z} + \Gamma v_1^+ e^{+j\beta z}$$

$$= v_1^+ e^{-j\beta z} (1 + \Gamma e^{+j\beta z})$$

This is essentially standing wave

Also, $\Rightarrow |e^{j\theta}| = \cos\theta + j\sin\theta$

$$\Rightarrow |e^{+j\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

Maximum Value

$$= |v_1^+| (1 + |\Gamma|)$$

Minimum Value

$$= |v_1^+| (1 - |\Gamma|)$$

$$\Rightarrow VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

* Load Impedance

$$Z_L = \left[Z_0 \left(\frac{1 - |\Gamma|}{1 + |\Gamma|} \right) \right]$$

(50Ω)

Ideal Case

$Z_L = \text{Unknown Impedance}$

$Z_0 = \text{Terminated impedance}$

VSWR

Reflection Coefficient
(Γ)

Reflected Power

1

0

0

2

0.33

11.1%

∞

1

100%

